

RNNs

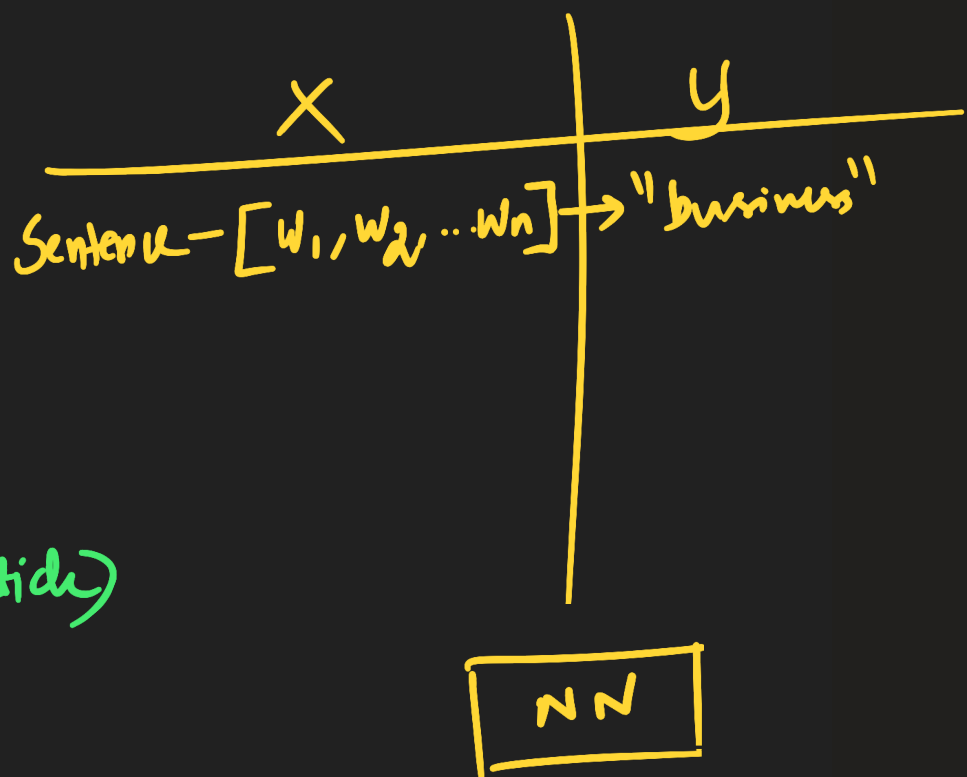
Satya Pattnaik

$$y = f(x)$$

↑ category

↑ text (news article)

Model



Why MLP is not meant for this problem?



	dog	bites	man
S1: dog bites man			
S2: man bites dog			

}

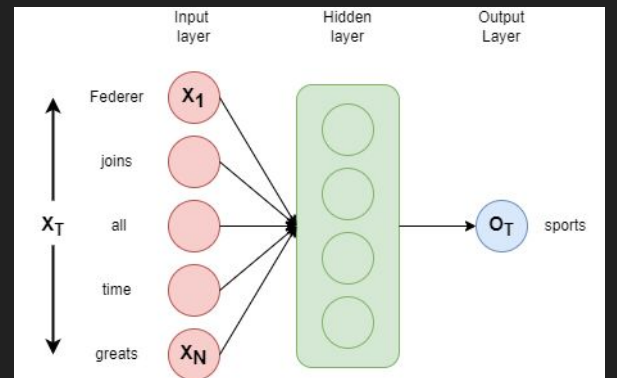
dog
bites
man

100 150 200 →

200 150 100

What if we use MLP?

All inputs at once



✓ not → -ve

✓ I ✓ love ✓ food

✓ I love good food ✓

← Sentence of 4 words

I am not a bad singer ✓

→ This is not a bad movie ✓

What is missing in this Architecture?

100 200 300 → +vec

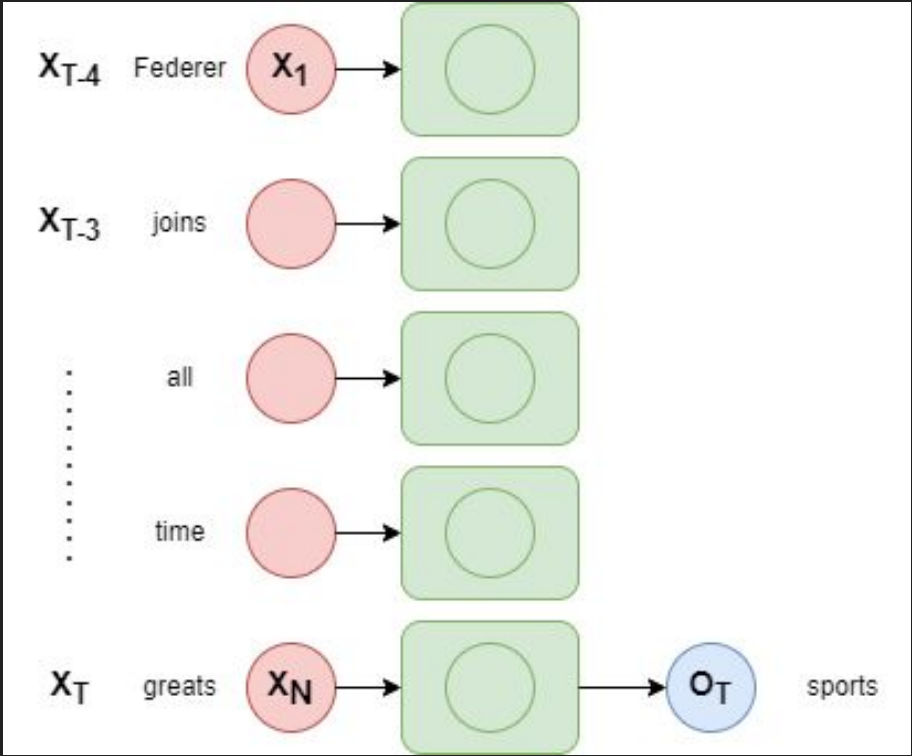
→ no connection with previous input words

200
300
100

→ each word is isolated from each other

→ varying lengths of sentences also will have issues

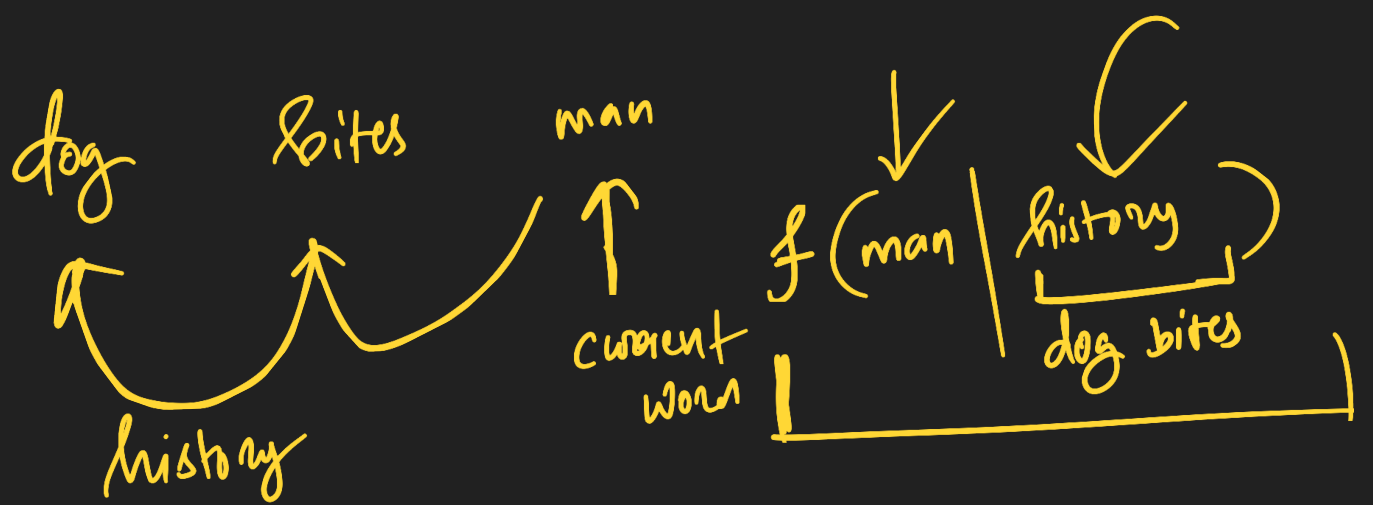
Can we modify this architecture?



What is missing?

Summarize the issues

$\begin{cases} \text{dog bites man} \\ \text{man bites dog} \end{cases}$ NN
 $ot = f(\text{word})$



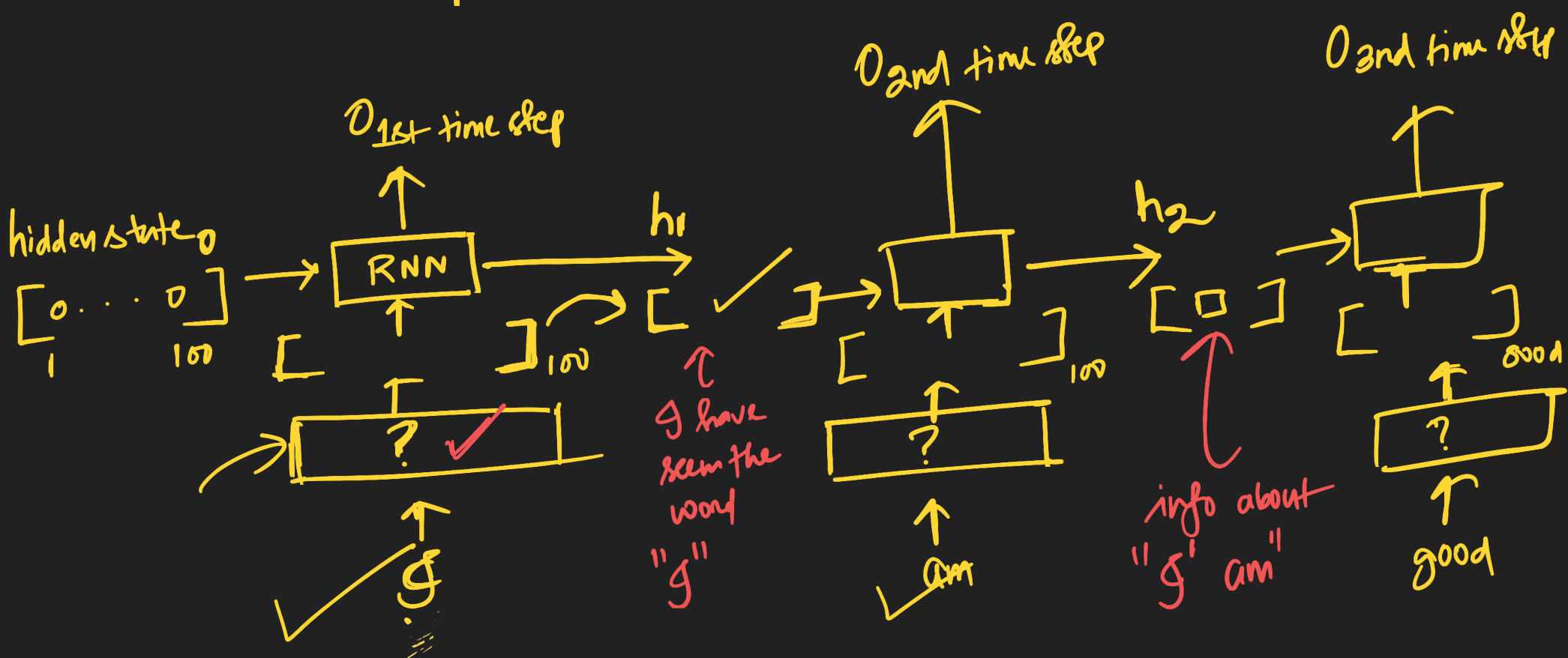
RNNs -> Intro

I
↑

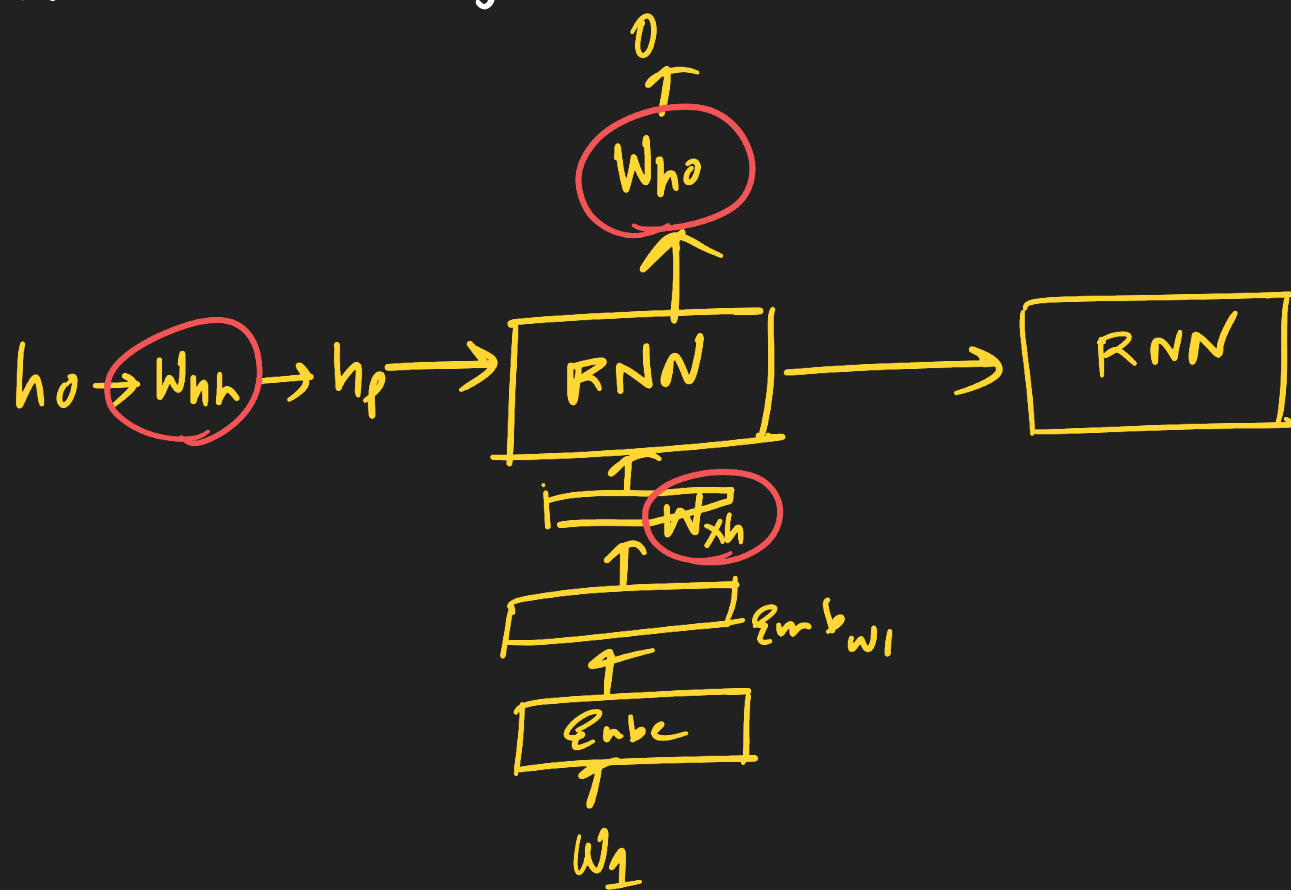
am

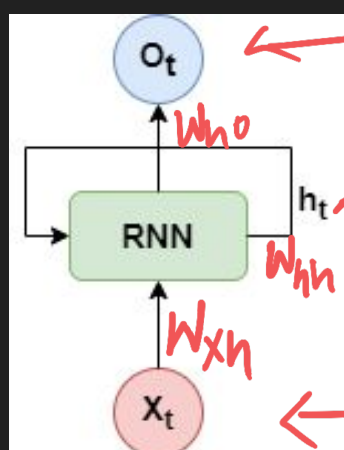
good

10 20 30 ^{stock}
↑ ↑ ↑
up



RNNs -> Advantages





output

hidden state
(captures history)

input word

How is it unravelled over time?

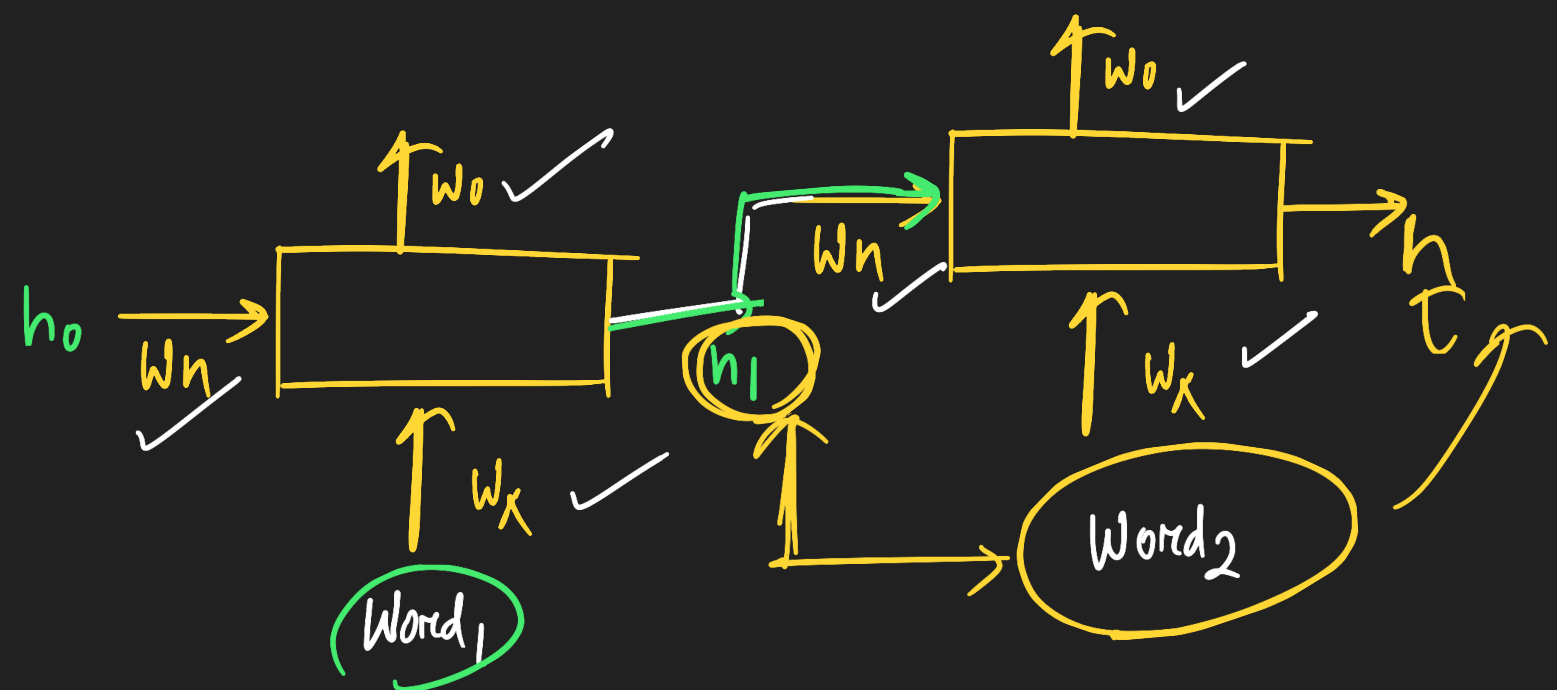
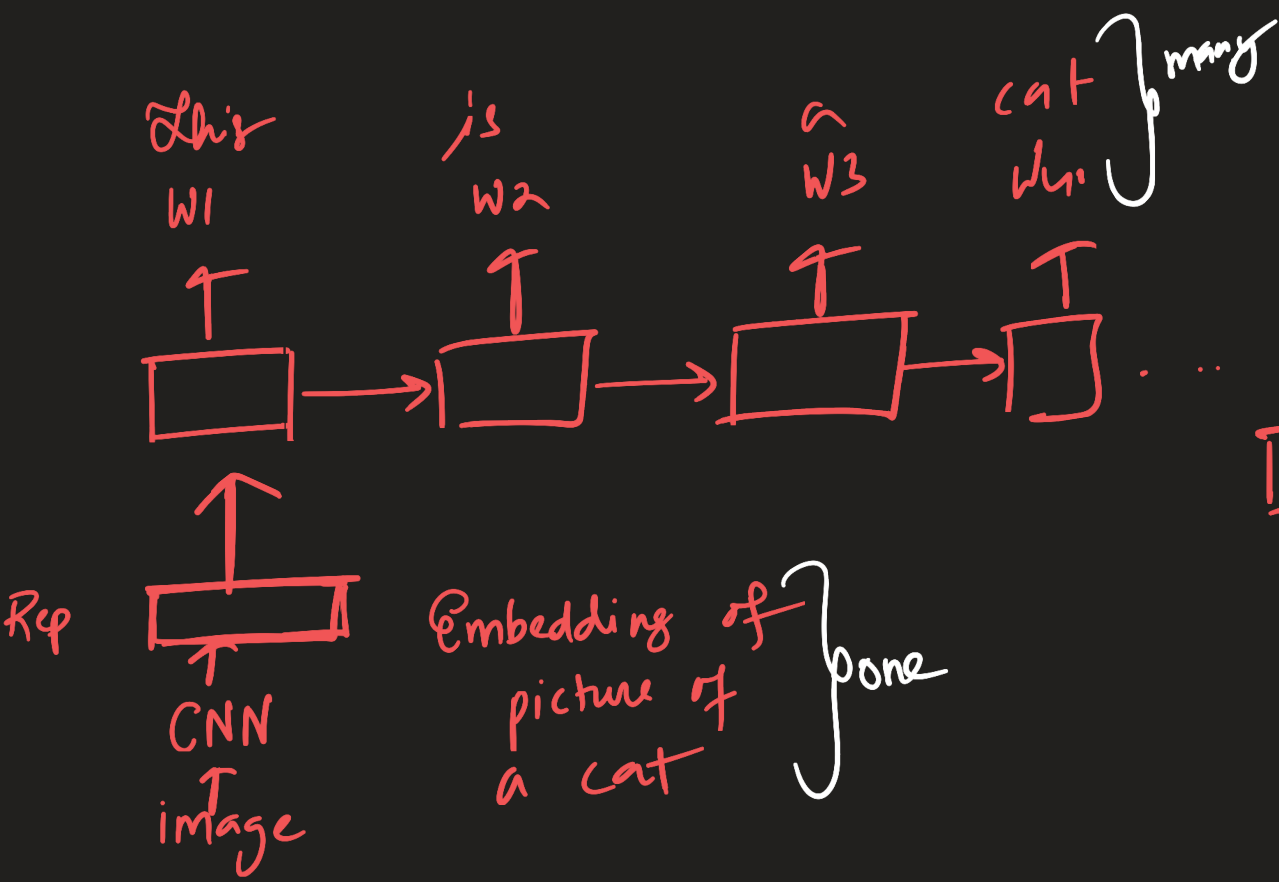
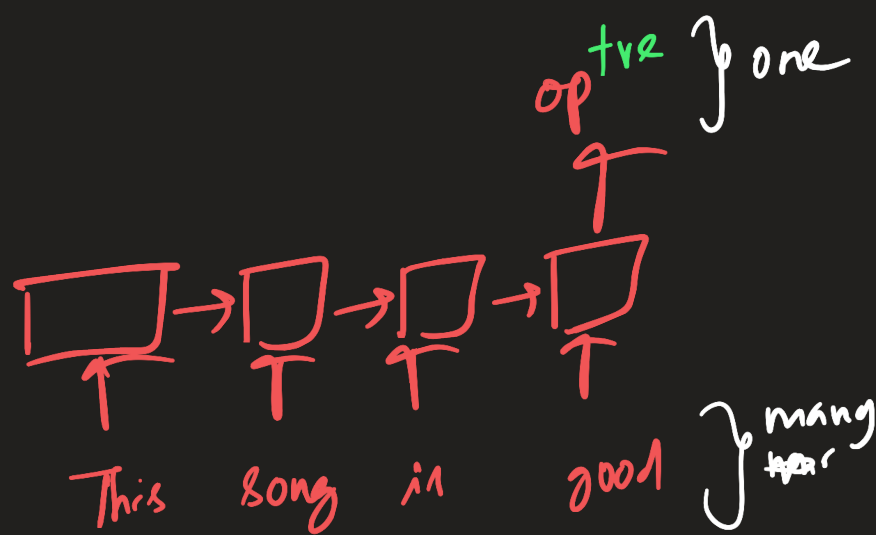
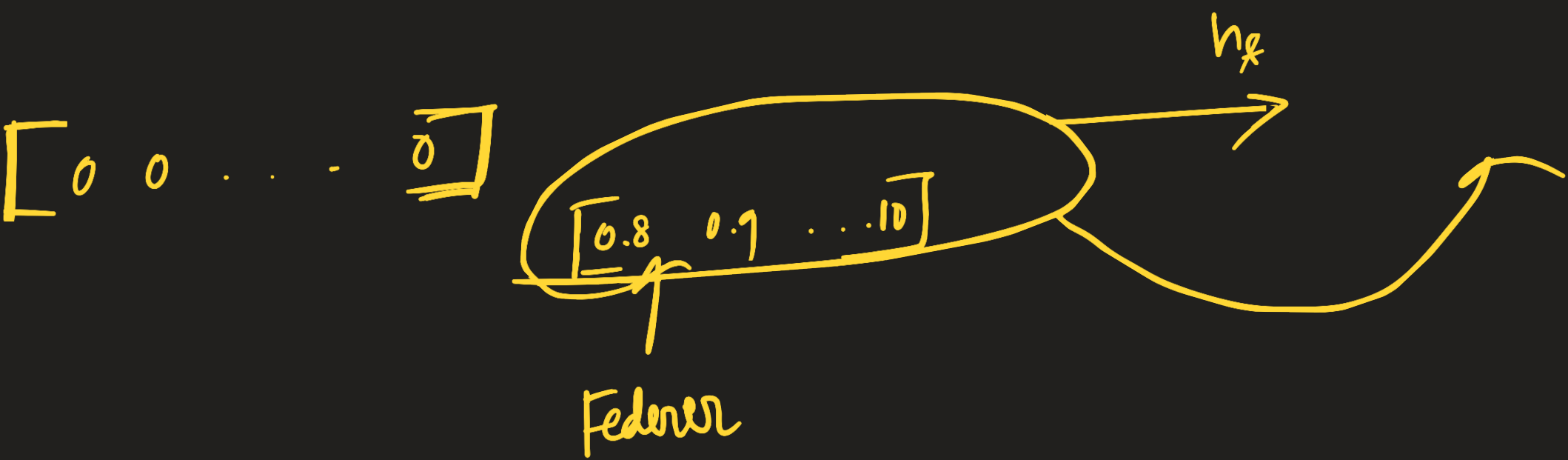
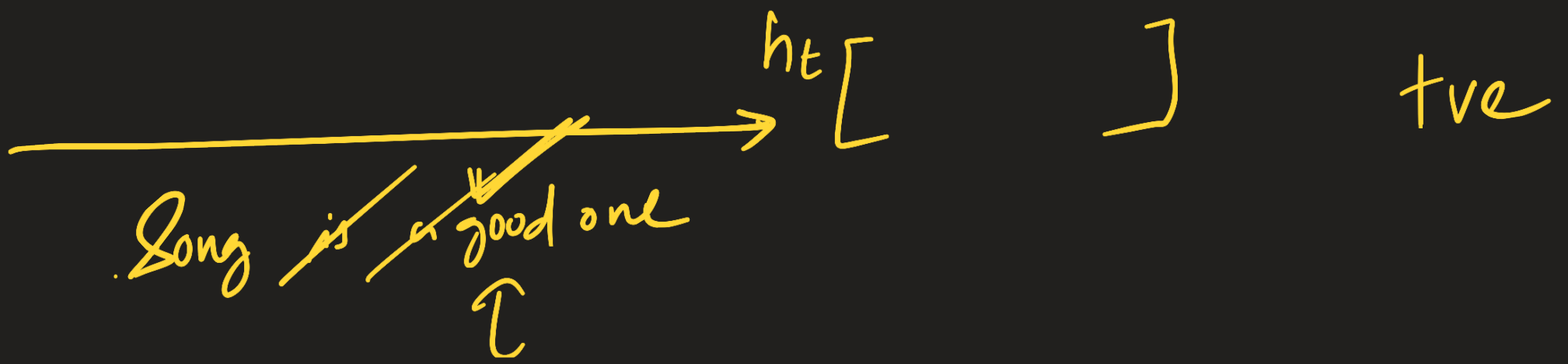


Image captioning



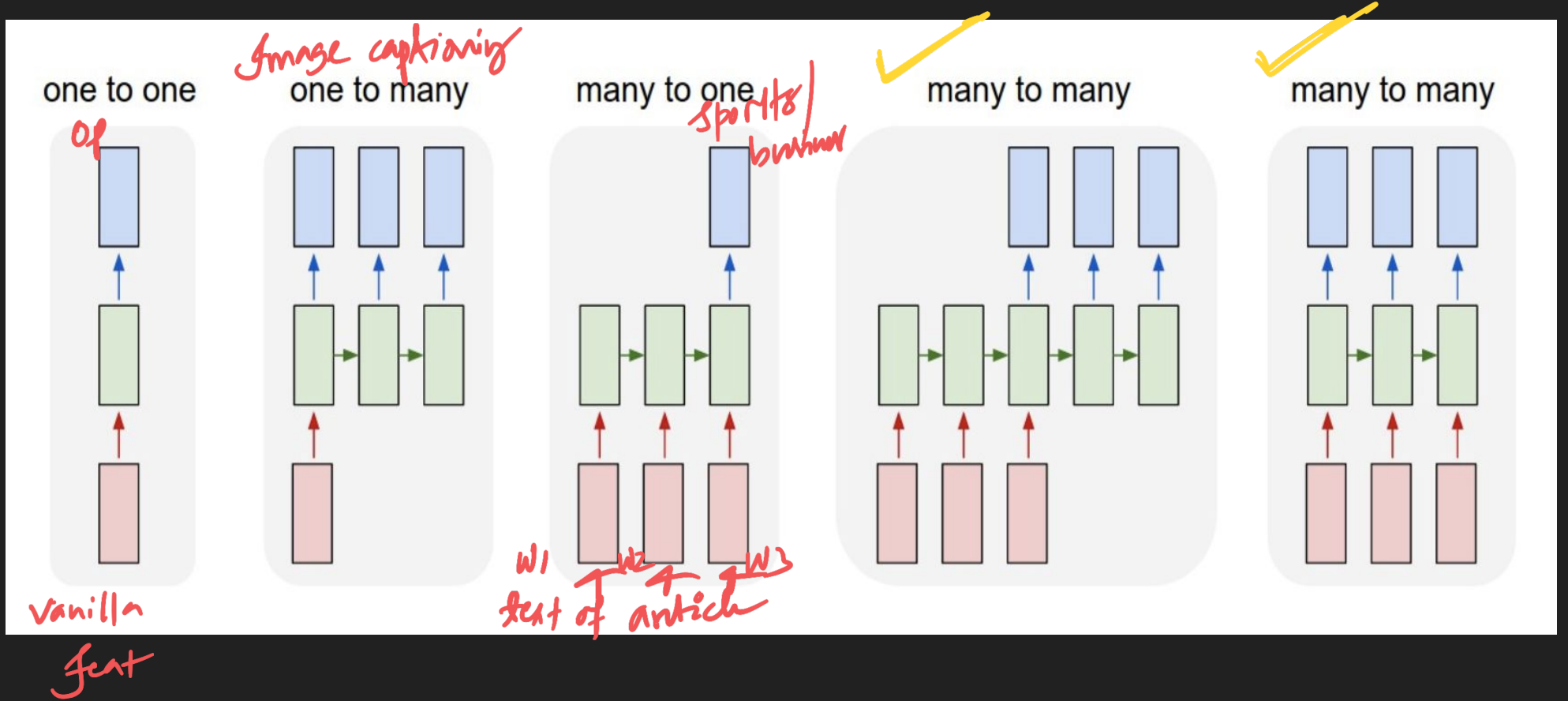
Sentiment classification





$$w_1 \quad w_2 \quad \rightarrow h_2 = f(\text{emb}_1, \text{emb}_2)$$

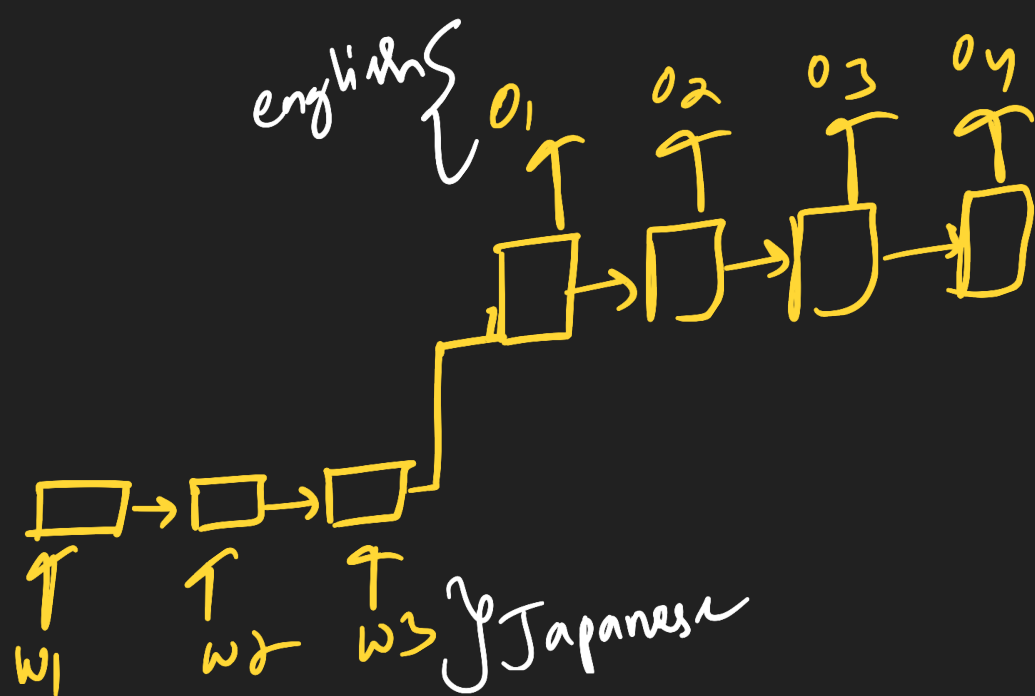
Types of RNN



Types of RNN

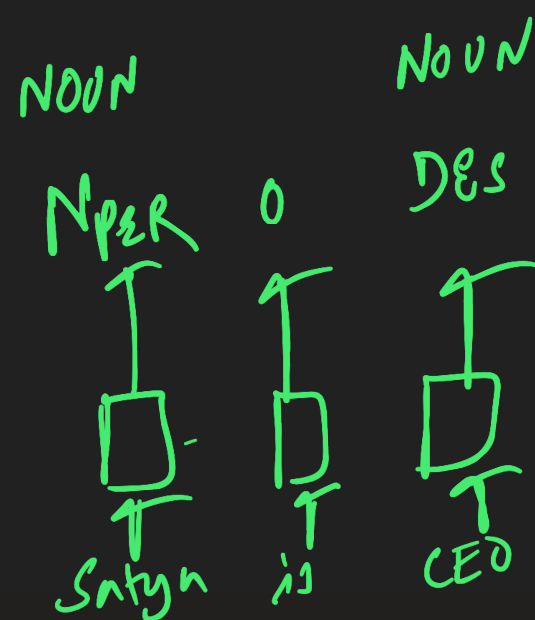
Many to Many

no. of ip $\neq$ no. of op



Many to Many ✓

no. of ip = no. of op



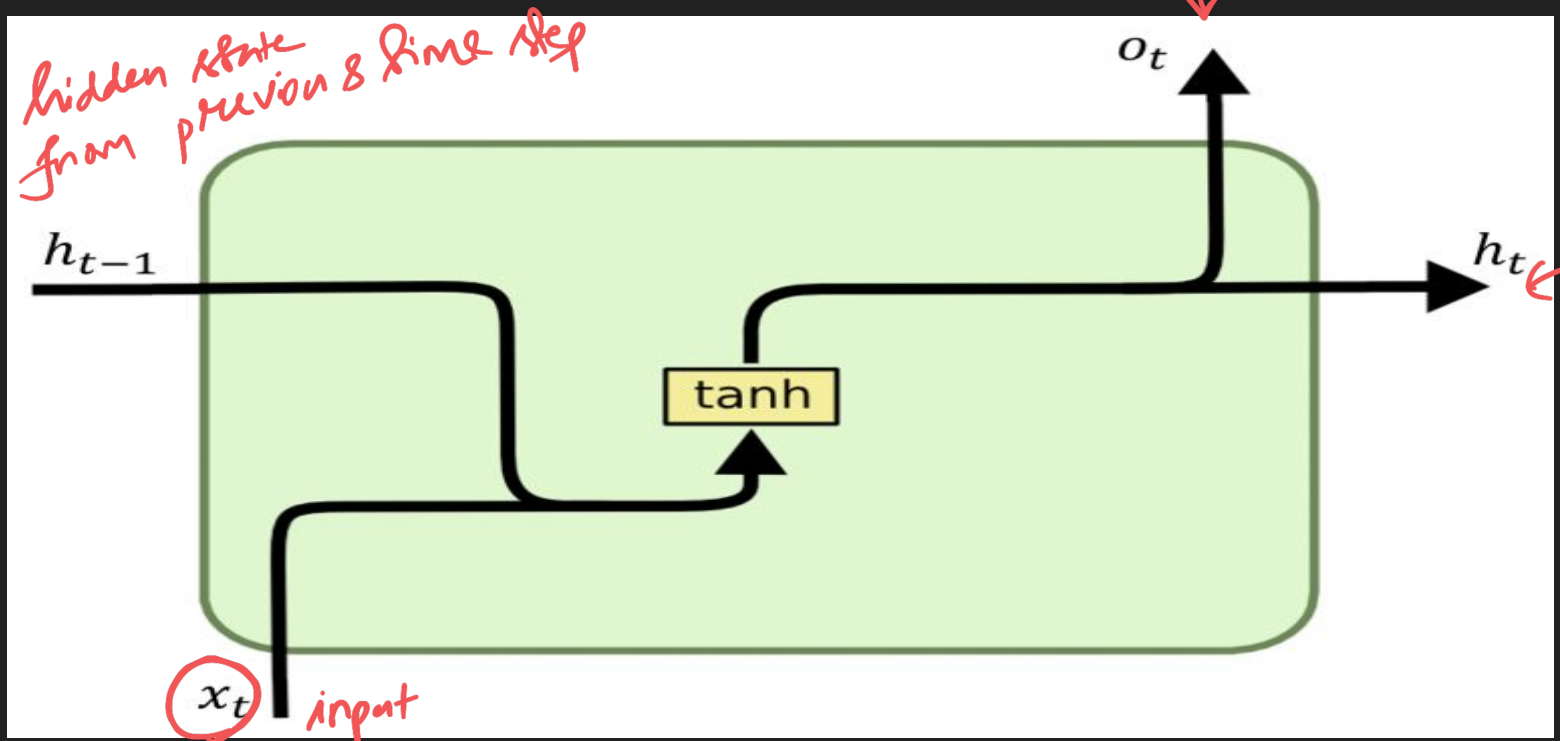
h_{t-1}
 100 200 $\rightarrow$

x_t
 300 $\uparrow$

$\left. \begin{array}{c} x_t \\ 300 \end{array} \right\} \begin{array}{l} o_t \\ h_t \end{array}$

$$\text{output} = Z(n_t)$$

$$h_t = F(x_t, h_{t-1})$$



hidden
state
for
next
time
step

But do we pass the words as it is to RNN cell?

Eiffel tower is great. where is it ?

True

But do we pass the words as it is to RNN cell?

1] Pre-trained embedding

2] Trainable embedding layer

WordVec
↑
g

Wav
↑
am

Wav
↑
good

vocab { g → 1
am → 2
good → 3

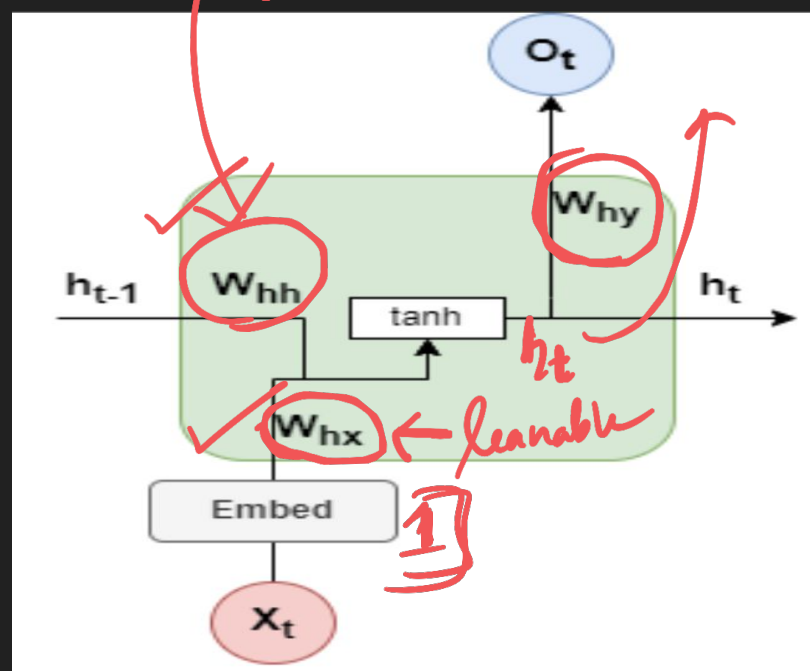
1 2 ... 100

Dense
g 1 0 0
am 0 0 0
good 0 0 0
W_{3x100}

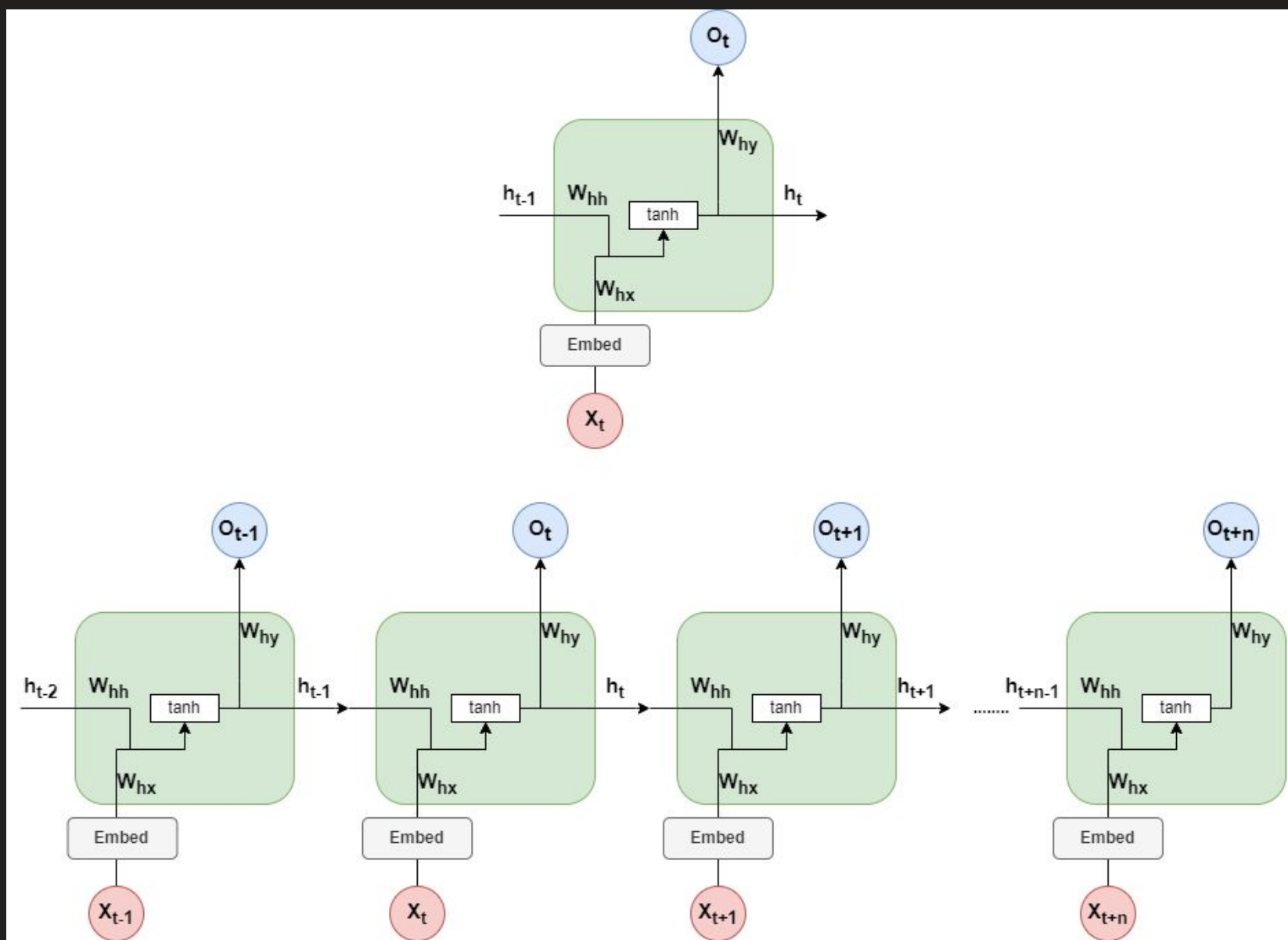
g 1 am 0 good 0
0 0 1 0
0 0 0 1

Weight is trainable

What is being learnt?

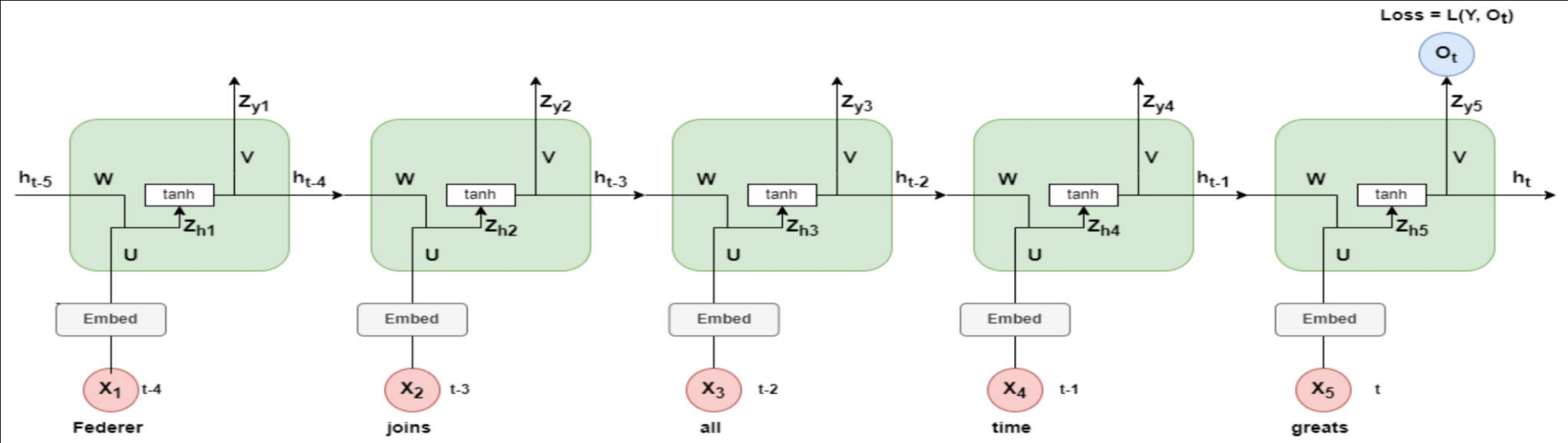


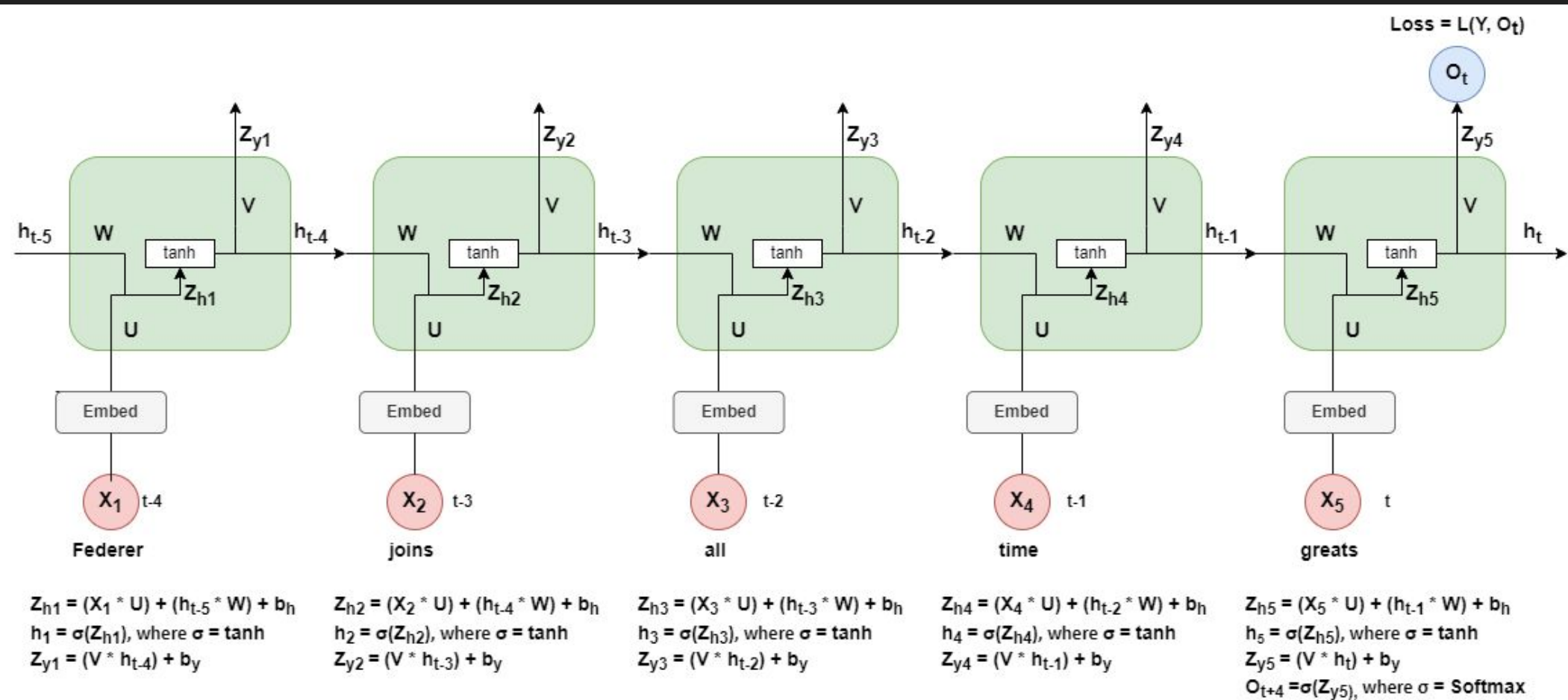
input word (embedding)



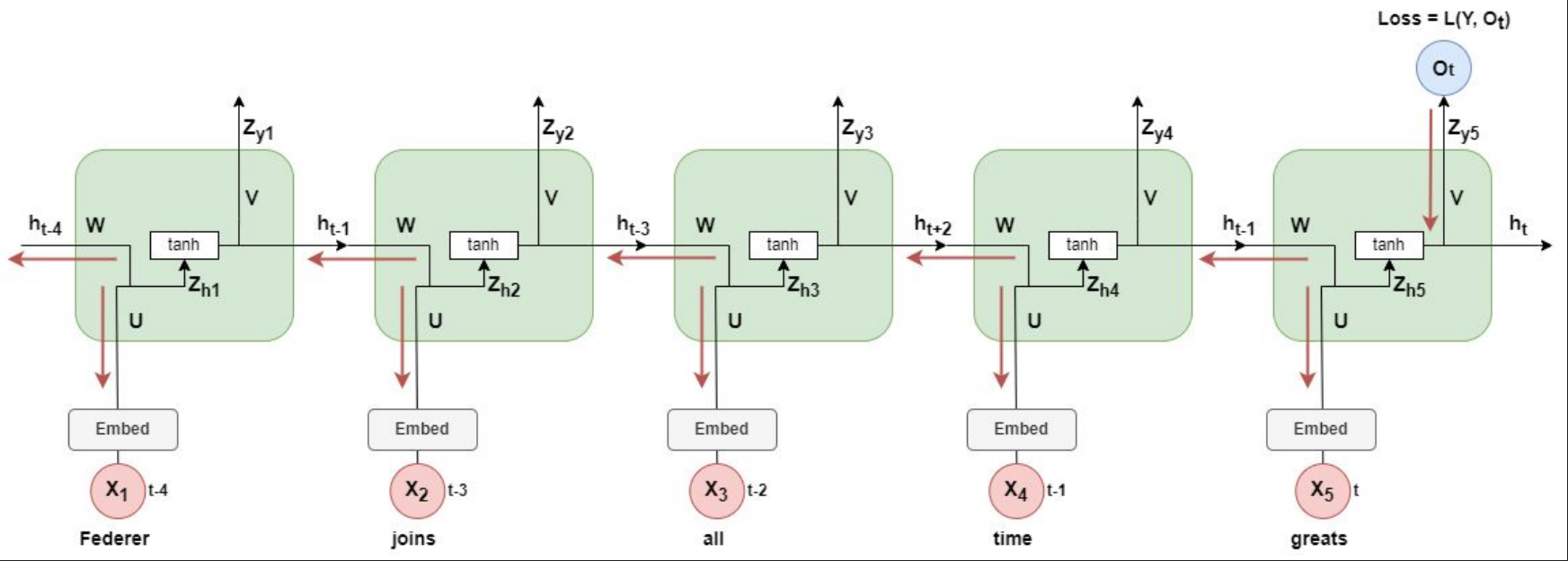
RNN Forward Pass

RNN Forward Pass

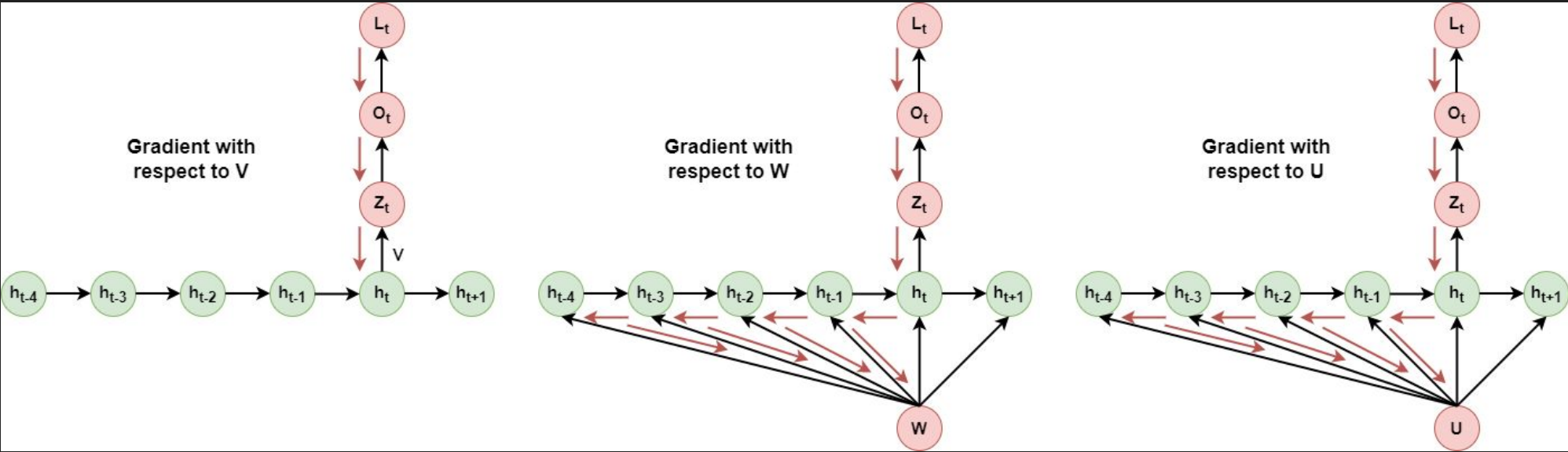




Steps



Weight Updates



Loss

$$\begin{aligned} \frac{\partial L_t}{\partial W} = & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial W} + \\ & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial W} + \\ & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial W} + \\ & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \frac{\partial h_{t-3}}{\partial W} + \\ & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \frac{\partial h_{t-3}}{\partial h_{t-4}} \frac{\partial h_{t-4}}{\partial W} + \\ & \frac{\partial L_t}{\partial O_t} \frac{\partial O_t}{\partial Z_t} \frac{\partial Z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \frac{\partial h_{t-3}}{\partial h_{t-4}} \frac{\partial h_{t-4}}{\partial h_{t-5}} \frac{\partial h_{t-5}}{\partial W} \end{aligned}$$

Let's generalize this to a single formula

$$\frac{\partial L_t}{\partial \mathbf{W}} = \frac{\partial L_t}{\partial \mathbf{O}_t} \frac{\partial \mathbf{O}_t}{\partial \mathbf{Z}_t} \frac{\partial \mathbf{Z}_t}{\partial \mathbf{h}_t} \sum_{r=1}^t \frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_r} \frac{\partial \mathbf{h}_r}{\partial \mathbf{W}}$$

Update the weight matrix $\mathbf{W}$

$$\mathbf{W} = \mathbf{W} - \alpha \frac{\partial L_t}{\partial \mathbf{W}}$$

